

OPERATOR HANDBOOK

Meercata v1.00

Complete Operations Guide
Windows IDS with Suricata and Npcap

Sanitized deployment profile

Suricata 8.0.6 | Service: Automatic | Mode: passive IDS | Capture: 203.0.113.74 | Rules: 45 files / 41,147 rules

Contents

Use the PDF bookmarks panel or this list to move between chapters.

1. Purpose and system overview	5
1.1 How the components work together	5
1.2 Suricata compared with Meercata	5
2. System requirements and validated baseline	6
2.1 Minimum requirements	6
2.2 Validated deployment baseline	6
2.3 Before the first run	6
3. Quick start - daily operations	7
3.1 Start Meercata	7
3.2 Check system health	7
3.3 Start monitoring	7
3.4 View evidence in a second terminal	7
4. Main menu - complete reference	8
4.1 Correct defaults for this deployment	8
5. Suricata control	9
5.1 IDS (sniff-only)	9
5.2 IPS (WinDivert inline)	9
5.3 Status	9
5.4 Monitor	9
6. Understanding live output	10
6.1 Reading one flow	10
6.2 Alert triage	10
7. Evidence and raw logs	11

7.1 Important locations	11
7.2 PowerShell views	11
7.3 Retention and integrity	11
8. Rules and YAML - safe operations	12
8.1 Rule-files submenu	12
8.2 Rule-change procedure	12
8.3 YAML/NFQ submenu on Windows	12
8.4 CLI rule commands	12
9. Command-line reference	13
9.1 Explicit monitor command	13
10. Windows service operations	14
10.1 Status, start, stop, restart	14
10.2 Auto-start	14
10.3 When the capture IP changes	14
10.4 Do not uninstall the service for routine troubleshooting	14
11. Requirement checks and diagnostics	15
11.1 When to run Check requirements	15
11.2 What is checked	15
11.3 Known benign warnings	15
12. Troubleshooting	16
12.1 Three-step test when there are no events	16
12.2 Check engine errors	16
13. Maintenance and change control	17
13.1 Weekly checklist	17
13.2 After a rule/config update	17
13.3 Rule-source updates	17
13.4 Known backup	17

14. Operational security	18
14.1 Initial alert response	18
15. Operator cheat sheet	19
15.1 Launch and monitor	19
15.2 Service and logs	19
15.3 Healthy indicators	19
15.4 Stop behavior	19
Appendix A - Glossary	20
Appendix B - Source basis and version notes	21
Minimum support bundle	21

1. Purpose and system overview

This manual explains how to operate a configured Meercata v1.00 deployment on Windows. It covers daily operations, the interactive menu, monitoring, evidence, rule management, Windows services, command-line use, maintenance, and troubleshooting.

Current scope

This deployment runs as a passive IDS. It detects and records traffic but does not block connections. IPS requires a Suricata build with WinDivert support; the installed 8.0.6 binary reports that capability as disabled.

1.1 How the components work together

Npcap	Suricata	eve.json	Meercata
packet capture	IDS engine	raw events	monitor and evidence

Windows network traffic -> detection -> logs -> readable operator evidence

- Npcap captures packets from the Windows network adapter.
- Suricata analyzes packets with ET Open rules and writes events to eve.json.
- Meercata Monitor follows new events, displays FLOW/DNS/TLS/ALERT records, and writes alerts or actual drops to Trigger Drop-Alert.txt.
- The Suricata Windows service continues running in the background after the Meercata UI is closed.

1.2 Suricata compared with Meercata

Component	Responsibility	Must remain open?
Suricata service	Capture, detection, and raw logging	Yes; the Automatic service manages it
Meercata UI	Status, configuration, rules, and operator workflow	No
Meercata Monitor	Readable live events and evidence collection	Only while collecting new evidence

2. System requirements and validated baseline

2.1 Minimum requirements

Requirement	Minimum / purpose	Validated installation
Windows	Windows 10/11 or Windows Server	Available
Python	Python 3.10 or newer	Python 3.13
Suricata	Windows IDS engine with Npcap support	8.0.6 RELEASE
Npcap	Packet-capture driver	Installed and running
Administrator	Capture, service, and Program Files changes	Required for elevated operations
Rules	At least one valid .rules file	ET Open, 45 active files
Memory	Depends on the ruleset	About 1 GB observed
Active interface	Existing adapter IP or Npcap device	203.0.113.74 in examples

2.2 Validated deployment baseline

- Suricata Windows service: Running, Automatic, LocalSystem.
- Capture backend: Npcap/pcap on Wi-Fi address 203.0.113.74.
- Configuration: C:\Program Files\Suricata\suricata.yaml.
- Versioned rules: C:\Program Files\Suricata\Meercata v1.00\rules\et-open-8.0.6-20260816.
- Startup validation: exit code 0 with no Suricata error lines.
- Engine startup: 45 rule files processed and 41,147 rules loaded.
- Meercata regression tests: 50 passed, 2 skipped after the Windows monitor-launcher fix.

The capture IP can change

203.0.113.74 is a documentation address representing the current capture target. DHCP, Wi-Fi changes, VPN use, or a network reset can change the real address. Follow Chapter 12 if the service stops or packets disappear after a network change.

2.3 Before the first run

Meercata controls Suricata; it does not install the engine or packet-capture driver automatically.

- Install Python 3.10+ and enable Add Python to PATH. Normal use needs no third-party pip packages.
- Install Npcap and confirm the driver is running. WinPcap API-compatible mode is recommended for service compatibility.
- Install Suricata for Windows with suricata.exe, readable YAML, valid rules, and EVE JSON enabled.
- Extract Meercata v1.00, run meercata-admin.cmd, select 7 Check requirements, and set the active adapter IP.

Mode summary

IDS = Suricata + Npcap. IPS = IDS requirements plus a WinDivert-enabled Suricata build. Monitor = a readable eve.json that Suricata is actively updating.

3. Quick start - daily operations

3.1 Start Meercata

From PowerShell in the outer folder:

```
PS C:\Program Files\Suricata\Meercata v1.00>  
& ".\Meercata v1.00\meercata-admin.cmd"
```

1. Approve the UAC prompt.
2. Confirm that the header shows config [ok], interface (ok), and service running.
3. Choose the menu for the task. Do not start foreground IDS for routine monitoring while the service already captures traffic.

3.2 Check system health

From the main menu choose 1) Suricata control, then 3) Status. Healthy output resembles:

```
[Suricata service] running  
[Processes]  
9376 suricata.exe  
[Capture] Npcap/pcap (Windows); Linux NFQUEUE counters do not apply
```

The PID changes

A process ID such as 9376 changes after every service restart. The important indicators are service=running and at least one suricata.exe process.

3.3 Start monitoring

1. From Suricata control choose 4) Monitor.
2. At EVE log path, press Enter to accept C:\Program Files\Suricata\log\eve.json.
3. At evidence path, press Enter to accept .\Trigger Drop-Alert.txt.
4. Leave the terminal open. Press Ctrl+C to stop only the monitor.

```
[INFO] Evidence log: C:\Program Files\Suricata\Meercata v1.00\  
Meercata v1.00\Trigger Drop-Alert.txt  
[INFO] EVE monitoring and evidence collection will continue until Ctrl+C.
```

3.4 View evidence in a second terminal

```
Get-Content "C:\Program Files\Suricata\Meercata v1.00\  
Meercata v1.00\Trigger Drop-Alert.txt" -Tail 100 -Wait
```

4. Main menu - complete reference

Option	Function	Windows guidance
1 Suricata control	IDS, IPS, Status, Monitor	Use Status and Monitor for routine operations
2 Threat Monitoring	Show the last 100 evidence lines and follow new records	Requires the Monitor collector in another terminal
3 Rules / YAML	View/toggle rules, add a rule, edit YAML	Use carefully; restart the service after changes
4 Hooks / firewall	Linux NFQUEUE firewall hooks	Unavailable on Windows
5 Systemd unit	Linux service lifecycle	Unavailable on Windows
6 Set defaults	Save config, interface, queue, backend	Use when the IP/interface changes
7 Check requirements	Complete preflight	Run after install, update, or network change
0 Quit	Close the Meercata UI	The Suricata service keeps running

4.1 Correct defaults for this deployment

Field	Value
Suricata config	C:\Program Files\Suricata\suricata.yaml
Interface	203.0.113.74
Queue	0 (not used by Windows IDS)
Backend	auto

Meercata defaults are stored in the Windows user profile. Changing the default interface in the UI does not automatically modify the Windows service command line; see Chapter 10 for service changes.

5. Suricata control

5.1 IDS (sniff-only)

The IDS menu runs Suricata in the foreground. By default it displays a dry run. If Apply is selected, Meercata can stop the existing service, start foreground capture, and attempt to restore the service when you exit with Ctrl+C.

Operational recommendation

For this machine, use the Windows service for 24/7 capture. Use foreground IDS only for troubleshooting or interface testing. Avoid running two Suricata capture instances on the same interface.

5.2 IPS (WinDivert inline)

IPS requires a Suricata binary built with WinDivert enabled: yes. The installed binary reports WinDivert disabled, so the requirement check rejects this option. Passive IDS remains fully functional.

5.3 Status

Status is read-only. It reports the service, processes, and capture backend. Use it after reboot, service restart, rule changes, or when the monitor receives no events.

5.4 Monitor

Monitor follows only new events written after it starts. On Windows it opens eve.json at the end, builds an in-memory DNS/domain cache, displays events, and writes only alerts or actual drops to the evidence file.

- Ctrl+C stops the monitor, not the Suricata service.
- The DNS/domain cache resets when the monitor restarts.
- Type off at the evidence prompt for display-only operation.
- Never use eve.json as the evidence-file destination.

6. Understanding live output

Label	Meaning	Operator action
FLOW	Observed network connection or flow	Normally benign; review IP, port, and protocol when unusual
DNS	DNS request or response	Use the domain for destination context
TLS	TLS metadata such as SNI/certificate	Review hostname, issuer, and destination
HTTP	HTTP host/request metadata	Review host, method, URL, and status
STATS	Periodic engine statistics	Confirms the engine is still processing
ALERT	A detection rule matched	Triage signature, priority, source, destination, and context
DROP ALERT	An alert that was actually blocked/dropped	Not expected in the current passive IDS mode

6.1 Reading one flow

```
2026-01-01T12:00:00+0800 [FLOW]
203.0.113.74:63367 -> 203.0.113.254:53 proto=UDP action=-
```

- 203.0.113.74 represents the monitored local host.
- 203.0.113.254:53 represents the DNS server/gateway on UDP port 53.
- action=- means the event contains no explicit block/drop action.
- A FLOW record alone is not proof of a threat. Use ALERT, domain, process context, and organizational baselines.

6.2 Alert triage

1. Record event time, signature, category, priority, and Rule ID (GID:SID:REV).
2. Identify source and destination IP/port and determine which host is internal.
3. Review DNS/domain, TLS SNI, HTTP host, application protocol, and interface.
4. Distinguish allowed traffic from an actual block. Passive IDS normally records only.
5. Find related eve.json events by timestamp, flow_id, IP, or SID.
6. Corroborate with endpoint, DNS, and browser/process logs before escalation.
7. Preserve or hash evidence when chain of custody matters.

Priority

In Suricata rules, a smaller priority number is usually more important (for example, 1 is higher than 3). Decisions must still consider asset value, context, recurrence, and false-positive likelihood.

7. Evidence and raw logs

7.1 Important locations

File / folder	Contents
C:\Program Files\Suricata\log\eve.json	Raw JSON events: flow, DNS, TLS, HTTP, alerts, stats, and more
C:\Program Files\Suricata\log\suricata.log	Engine startup, rule loading, warnings, and errors
C:\Program Files\Suricata\log\fast.log	One-line alert summary
C:\Program Files\Suricata\log\stats.log	Packet, flow, decoder, memory, and application-layer statistics
...\Meercata v1.00\Trigger Drop-Alert.txt	Readable evidence produced by Meercata Monitor
C:\Program Files\Suricata\suricata.yaml	Active Suricata configuration

7.2 PowerShell views

```
# Alert evidence
Get-Content ".\Trigger Drop-Alert.txt" -Tail 100 -Wait

# Engine log
Get-Content "C:\Program Files\Suricata\log\suricata.log" -Tail 80

# Fast alerts
Get-Content "C:\Program Files\Suricata\log\fast.log" -Tail 50 -Wait
```

7.3 Retention and integrity

- eve.json can grow quickly. Monitor free space and use an approved rotation/retention policy.
- Trigger Drop-Alert.txt is an operational hunting log, not a tamper-proof forensic record.
- For incidents, copy evidence to protected read-only storage and record its hash and collection time.
- Do not edit eve.json while Suricata is writing to it.
- Do not publish raw logs without sanitization; they can contain IPs, hostnames, domains, URLs, certificates, and sensitive metadata.

8. Rules and YAML - safe operations

Before making changes

Confirm administrator rights, a configuration backup, an approved maintenance window when needed, and command validation. Valid YAML changes are not used by the service until Suricata reloads or restarts.

8.1 Rule-files submenu

Option	Function	Note
Show rule-files	List enabled/disabled entries	Read-only
Rule menu	Toggle a rule file interactively	YAML tools create a backup and validate
Rule set (manual)	Set one filename enabled/disabled	Name must match rule-files
Add rule-file	Create a local .rules file and add it to YAML	Insert only tested rule syntax

8.2 Rule-change procedure

1. Choose 3 Rules / YAML -> 1 Rule files -> 1 Show rule-files and record the current state.
2. Make only one change at a time.
3. Require Meercata/Suricata validation to pass. Never force a failed configuration.
4. Restart the service in elevated PowerShell: Restart-Service Suricata.
5. Check Status and suricata.log for Engine started and the expected rule count.
6. Monitor alert volume and roll back an unacceptable flood or false-positive spike.

8.3 YAML/NFQ submenu on Windows

- YAML summary shows HOME_NET, EXTERNAL_NET, capture settings, runmode, and rule path.
- af-packet is a Linux capture configuration; Windows live IDS uses pcap/Npcap.
- NFQ batchcount, fail-open, and queue are Linux NFQUEUE settings and do not control Windows Npcap IDS.
- workers/autofp runmode selection is advanced tuning; change it only after measurement and validation.

8.4 CLI rule commands

```
# View rules
.\meercata.cmd rules
# Disable one file - global --apply must precede the command
.\meercata.cmd --apply rule-set --file custom.rules --state disabled
# Re-enable
.\meercata.cmd --apply rule-set --file custom.rules --state enabled
# Validate the installed configuration
& "C:\Program Files\Suricata\suricata.exe" --init-errors-fatal -T `
-c "C:\Program Files\Suricata\suricata.yaml"
```

9. Command-line reference

For CLI use, open PowerShell as Administrator and enter the application folder. meercata-admin.cmd starts only the interactive UI and does not accept arguments.

```
Set-Location "C:\Program Files\Suricata\Meercata v1.00\Meercata v1.00"
```

Task	Command
Help	.\meercata.cmd --help
Requirement check	.\meercata.cmd check
Status	.\meercata.cmd status
Combined diagnostics	.\meercata.cmd diag
View rules	.\meercata.cmd rules
YAML summary	.\meercata.cmd yaml-summary
Dry-run foreground IDS	.\meercata.cmd --iface 203.0.113.74 ids
Apply foreground IDS	.\meercata.cmd --iface 203.0.113.74 --apply ids
Monitor	.\meercata.cmd monitor --log C:\Program Files\Suricata\log\eve.json

Argument order

Global options such as --config, --iface, --queue, --backend, and --apply must appear before the subcommand. Monitor-specific options such as --log and --trigger-log appear after monitor.

9.1 Explicit monitor command

```
.\meercata.cmd monitor `
--log "C:\Program Files\Suricata\log\eve.json" `
--trigger-log ".\Trigger Drop-Alert.txt"
```

10. Windows service operations

Every command in this chapter requires elevated PowerShell.

10.1 Status, start, stop, restart

```
Get-Service Suricata
Start-Service Suricata
Stop-Service Suricata
Restart-Service Suricata

# Native service details
sc.exe qc Suricata
sc.exe queryex Suricata
```

10.2 Auto-start

```
Set-Service -Name Suricata -StartupType Automatic
```

10.3 When the capture IP changes

1. Run ipconfig and identify the IPv4 address on the adapter with a Default Gateway.
2. In Meercata, choose menu 6 and save the new interface.
3. Validate the configuration.
4. Change Windows service parameters using the space-free configuration path below.
5. Restart the service and confirm that new events appear in eve.json.

```
# Replace NEW_IP with the active adapter IP
& "C:\Program Files\Suricata\suricata.exe" `
--service-change-params `
-c C:\PROGRA~1\Suricata\suricata.yaml `
-i NEW_IP
Restart-Service Suricata
```

Why PROGRA~1?

The Suricata 8.0.6 Windows service installer on this system did not preserve quotes around option values. The space-free 8.3 path prevents C:\Program Files from being split during service startup.

10.4 Do not uninstall the service for routine troubleshooting

Stopping the service or correcting the interface is normally sufficient. Service removal is a destructive but reversible change and should be used only when intentionally removing the deployment.

11. Requirement checks and diagnostics

11.1 When to run Check requirements

- After installing or upgrading Python, Suricata, Npcap, or rules.
- After changing Wi-Fi, Ethernet, VPN, Hyper-V, VMware, or the IP address.
- When config [ok] changes to missing or unreadable.
- When foreground IDS or Monitor fails to start.

```
.\meercata.cmd check  
.\meercata.cmd diag
```

11.2 What is checked

- Python version and launcher availability.
- Suricata executable, configuration path, and static build capabilities.
- Npcap/pcap capture availability and administrator restrictions.
- Interface presence and IP mapping.
- EVE log readability for monitoring.
- Command-specific privileges and dependencies.

11.3 Known benign warnings

- Missing optional threshold.config: no custom threshold/suppression file is loaded; validation can still pass.
- Some ET flowbits checked but not set: possible with a subset of active rules; the engine can still load.
- setrlimit unavailable on Windows: a platform limitation, not a startup failure.

12. Troubleshooting

Symptom	Common cause	Corrective action
service=stopped	IP changed, config error, or Npcap problem	Validate config; check suricata.log and ipconfig; update service parameters; restart
Monitor has no output	No capture, monitor started before new events, or wrong EVE path	Check Status; generate DNS traffic; check eve.json LastWriteTime/Length
No module named suricatamon	Old launcher started from System32	Exit every UI and relaunch after the launcher fix; use the current meercata.cmd
Evidence path becomes System32	Elevated working directory was not anchored	Exit and relaunch current meercata-admin.cmd; expected path is the application folder
Access denied	Terminal is not elevated or Npcap is admin-only	Use meercata-admin.cmd or Run as administrator
Config validation fails	YAML/rule syntax error or missing file	Do not restart; read the error; restore the timestamped backup
Many alerts	Noisy rule, new baseline, or real scan	Triage SID/context; do not disable a whole category without review
High memory	Large ruleset and flow state	Check stats.log, rule count, and traffic; tune only after baselining

12.1 Three-step test when there are no events

```
# 1. Service
Get-Service Suricata
# 2. Generate local traffic
ping.exe -n 4 203.0.113.254
Resolve-DnsName example.com
# 3. Confirm log movement
Get-Item "C:\Program Files\Suricata\log\eve.json" | `
  Select-Object LastWriteTime,Length
```

12.2 Check engine errors

```
Get-Content "C:\Program Files\Suricata\log\suricata.log" `
  -Tail 150 | Select-String "Error:|Emergency:| E:"
```

Do not delete logs to fix the engine

Logs are diagnostic evidence. Copy or rotate them using an approved procedure; do not delete them at random while the service is writing.

13. Maintenance and change control

13.1 Weekly checklist

- [] Suricata service remains Running and Automatic.
- [] The service interface/IP still matches the active adapter.
- [] eve.json and stats.log continue growing.
- [] Sufficient disk space remains.
- [] High-priority alerts and recurrence are reviewed.
- [] suricata.log is checked for new errors.
- [] Incident evidence is preserved in protected storage when required.

13.2 After a rule/config update

- [] A timestamped configuration backup exists.
- [] The new rule directory is versioned and complete.
- [] suricata -T exits 0 with no fatal error lines.
- [] Service restart succeeds.
- [] Engine started appears and the rule count is reasonable.
- [] Network connectivity remains normal.
- [] Alert volume is monitored for a false-positive spike.

13.3 Rule-source updates

Meercata v1.00 manages enabled/disabled entries and custom rule files, but it does not provide one-click Windows rules-feed updates. Use a versioned staging directory, configuration backup, pre-deployment validation, atomic replacement, installed validation, and rollback.

Current snapshot

The current ET Open snapshot is in et-open-8.0.6-20260816. Do not overwrite that versioned folder in place. Stage a new snapshot in a new folder, validate it, and then change the configuration in a controlled way.

13.4 Known backup

The successful deployment backup follows this pattern:

```
C:\Program Files\Suricata\suricata.yaml.YYYYMMDD-HHMMSS-NNN.meercata-rules.bak
```

14. Operational security

- Use least privilege. Elevate only for capture, configuration, and service actions that require it.
- Do not treat one alert as proof of compromise without corroboration.
- Do not disable broad rulesets merely to reduce noise; tune by SID and context.
- Never place credentials, private keys, or secrets in custom rules or evidence notes.
- Raw network metadata can be sensitive. Restrict access to logs and backups.
- Passive IDS does not replace endpoint protection, firewalls, patching, backups, or incident response.
- When accessing the host remotely, avoid network/service changes that could disconnect the session.

14.1 Initial alert response

1. Preserve evidence and record the collection time and location.
2. Identify the host, user, process, domain, and destination.
3. Review recurrence and other alerts on the same flow or IP.
4. Contain only according to organizational policy and authority.
5. Escalate with a factual summary, not assumptions.

15. Operator cheat sheet

15.1 Launch and monitor

```
# Interactive UI from the outer folder
& ".\Meercata v1.00\meercata-admin.cmd"

# Inside the UI
1 -> 3 Status
1 -> 4 Live EVE collector
2 Evidence viewer
3 Rules / YAML
6 Defaults
7 Requirements
```

15.2 Service and logs

```
Get-Service Suricata
Restart-Service Suricata
Get-Content "C:\Program Files\Suricata\log\suricata.log" -Tail 80
Get-Content "C:\Program Files\Suricata\log\fast.log" -Tail 50 -Wait
```

15.3 Healthy indicators

- service=running
- a suricata.exe process exists
- Engine started appears in suricata.log
- eve.json LastWriteTime and Length change
- FLOW/DNS/TLS/STATS records appear in Meercata Monitor
- no Error/Emergency startup line

15.4 Stop behavior

Action	What stops	What continues
Ctrl+C in Monitor	Meercata event collector	Suricata service and raw logging
Quit UI	Interactive menu	Suricata service
Stop-Service Suricata	Packet capture and detection	Meercata files; monitor waits for events
Windows shutdown	All processes	Service starts automatically at next boot

Appendix A - Glossary

Term	Meaning
IDS	Intrusion Detection System; detects and logs without inline blocking
IPS	Intrusion Prevention System; inspects inline and can block/drop
Npcap	Windows packet-capture driver and API
EVE JSON	Structured event format written by Suricata
Rule / signature	Detection logic with a SID, revision, category, and action
SID	Signature ID
GID	Generator ID
REV	Rule revision
Flow	One communication stream between endpoints
SNI	Hostname announced during TLS negotiation
HOME_NET	Network considered protected/local in the Suricata configuration
False positive	A rule matched activity that is actually benign or approved
WinDivert	Windows packet-interception driver/API required by certain IPS builds

Appendix B - Source basis and version notes

This manual was prepared from the following local files and validated deployment behavior:

- Meercata README.md and WINDOWS.md.
- Interactive menus and the CLI parser in main.py.
- Windows service/capture helpers in platform_support.py and suricata_ctl.py.
- Evidence formatter/follower in suricatamon.py.
- Live validation, service state, rule count, EVE output, and troubleshooting confirmed on 21-22 August 2026.

Version sensitivity

Commands and menus match Meercata v1.00 and the installed Suricata 8.0.6 release. Run the requirement check and review release notes after upgrades because menus, defaults, service syntax, and capabilities can change.

Minimum support bundle

When requesting technical support, include the following output after removing sensitive data:

```
.\meercata.cmd status
.\meercata.cmd diag
.\meercata.cmd check
Get-Content "C:\Program Files\Suricata\log\suricata.log" -Tail 150
sc.exe qc Suricata
sc.exe queryex Suricata
```

End of manual

Meercata v1.00 - Windows IDS Operations